

Practice Spring Boot APIs 8

Database Integration using Spring Data JPA

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Project Overview



REST API
APPLICATION



EMPLOYEE
MANAGEMENT



CAMPAIGN
MANAGEMENT



VEHICLE
MANAGEMENT



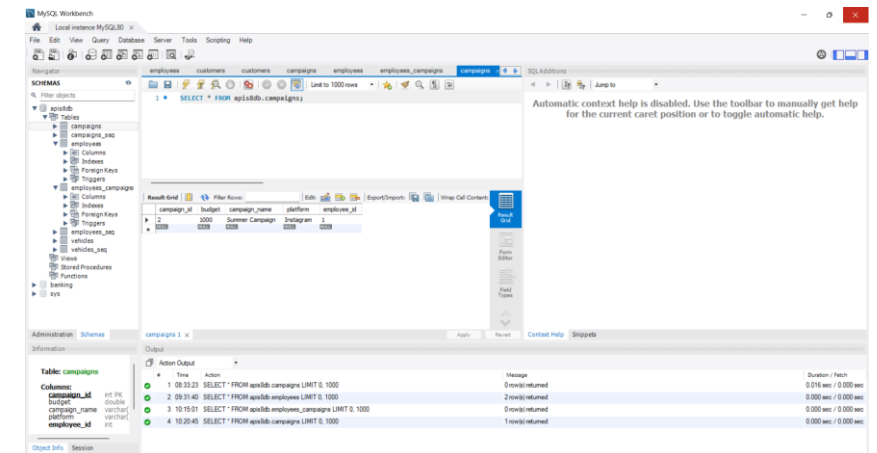
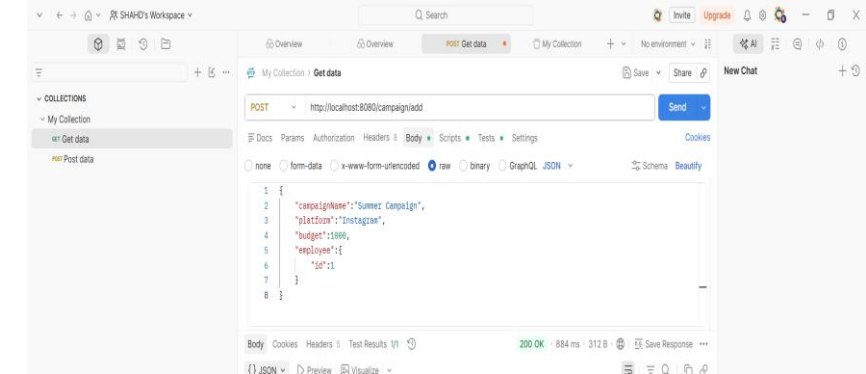
MYSQL
INTEGRATION

Database Design

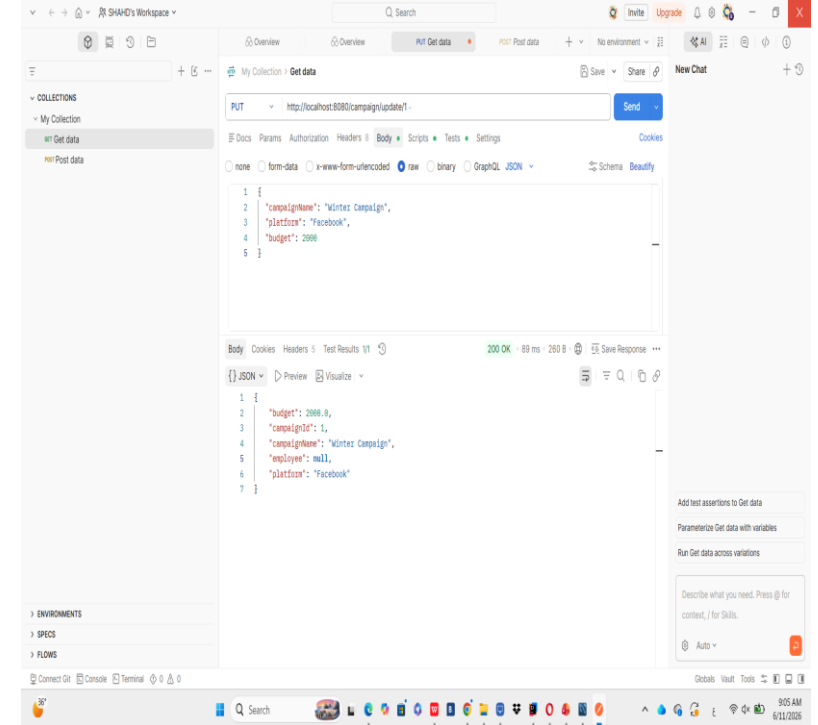
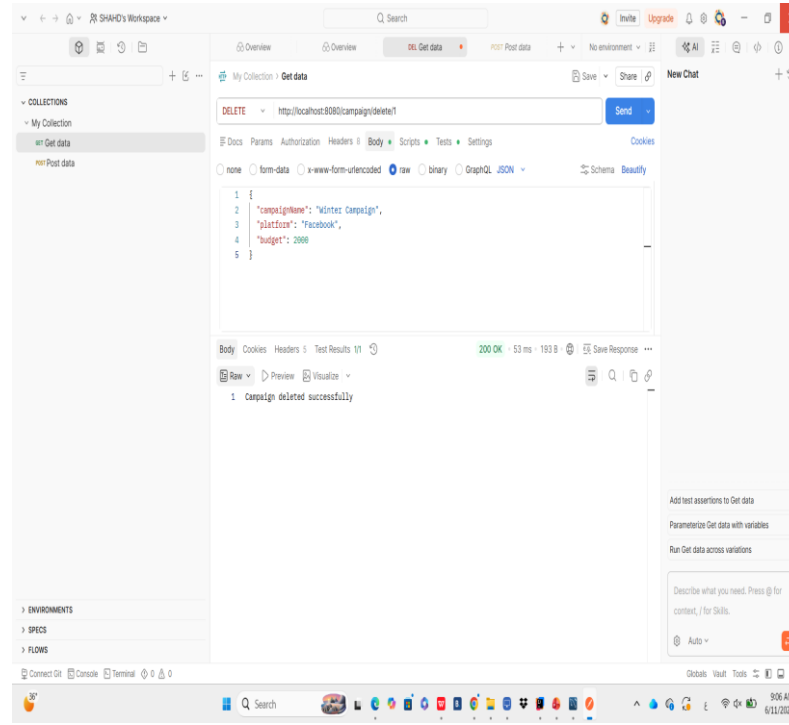
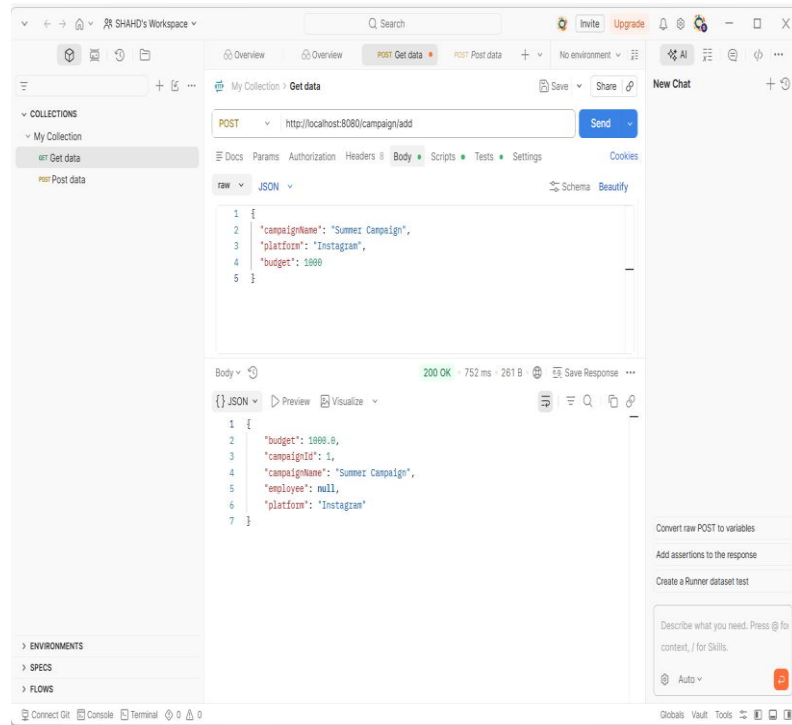
Entity	Main Fields
Employee	id, name, department
Campaign	campaignId, campaignName, platform, budget
Vehicle	vehicleId, vehicleModel, rentalPricePerDay

Entity Relationship

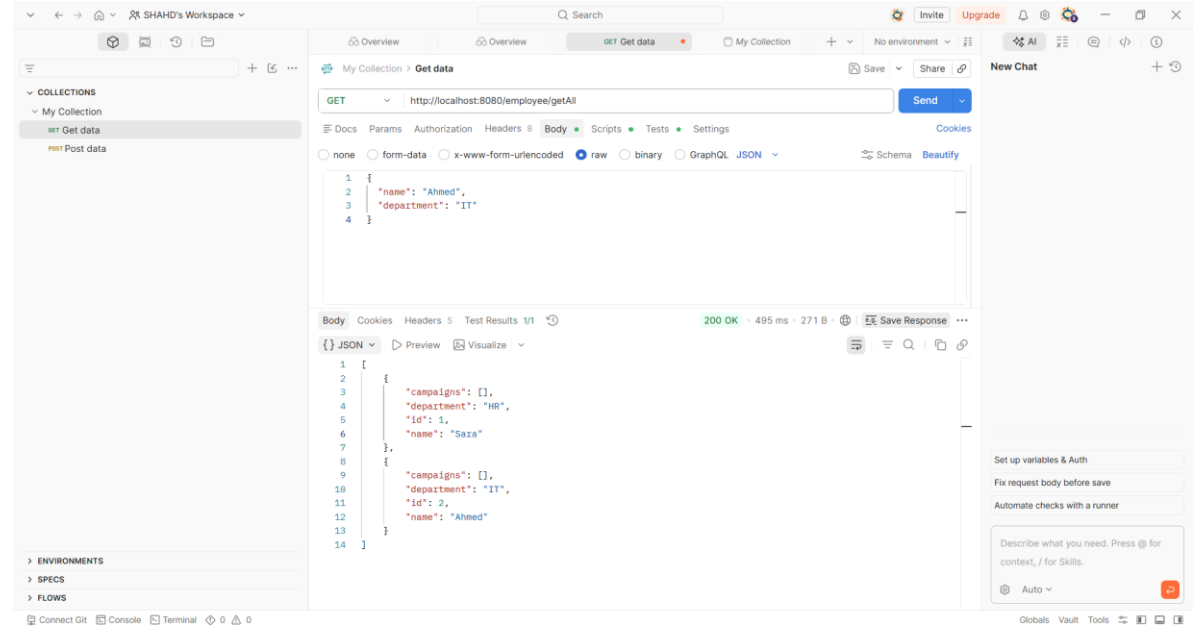
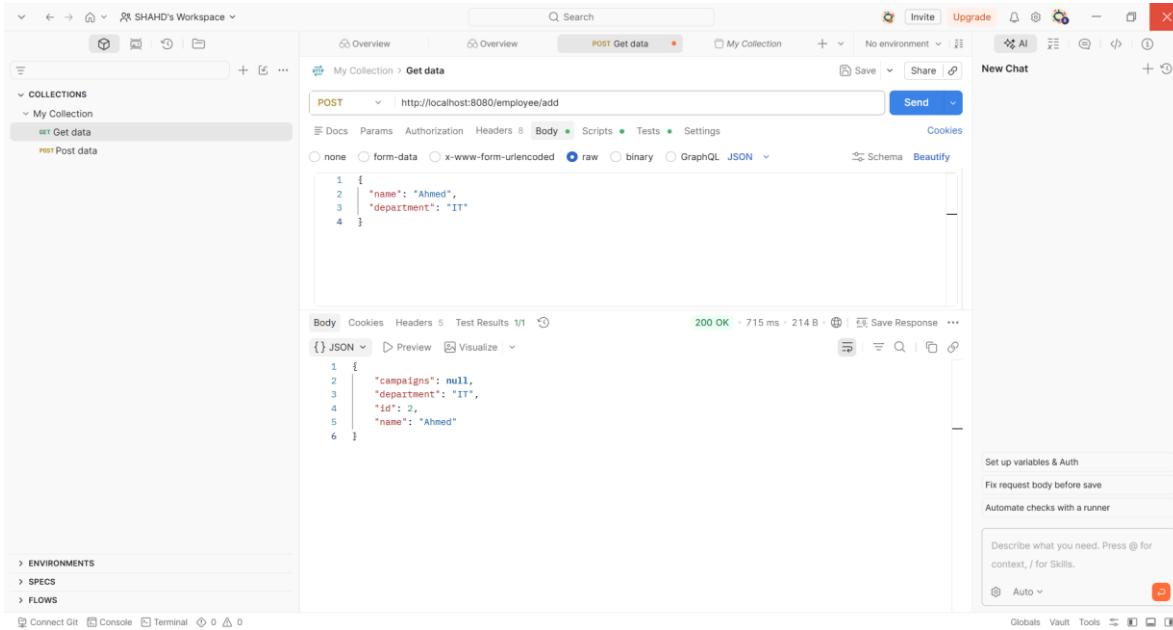
- Relationship Type
- Employee (1) -----> (Many) Campaigns
- @OneToMany(mappedBy = "employee")
- private List<Campaign> campaigns;
- @ManyToOne
- private Employee employee;
- **Description**
- One Employee can manage multiple Campaigns.
- Each Campaign belongs to one Employee



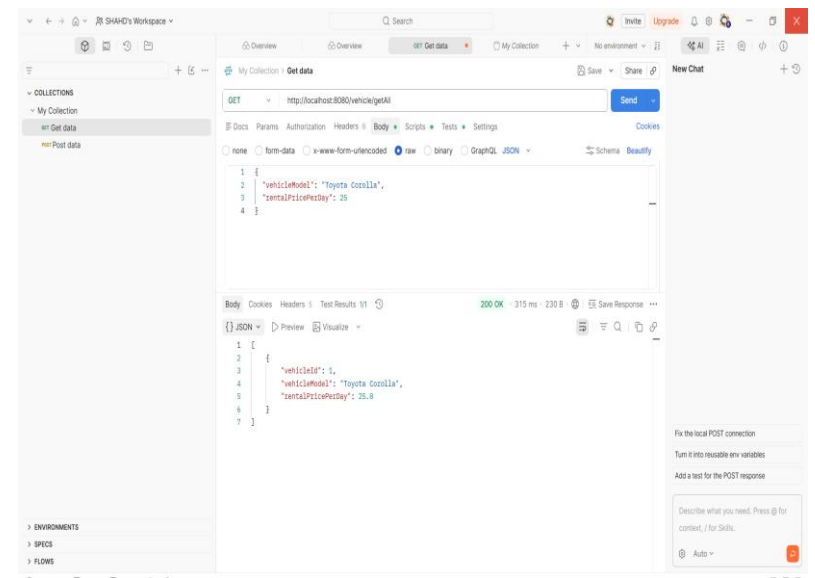
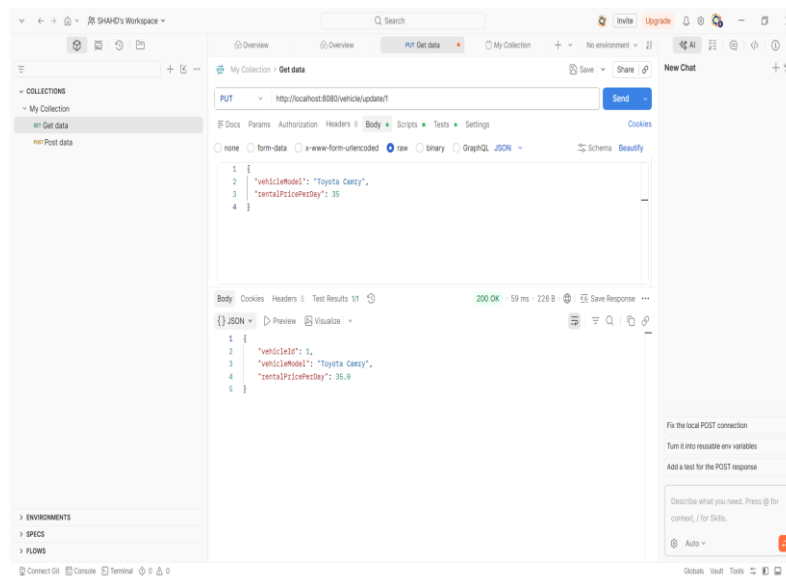
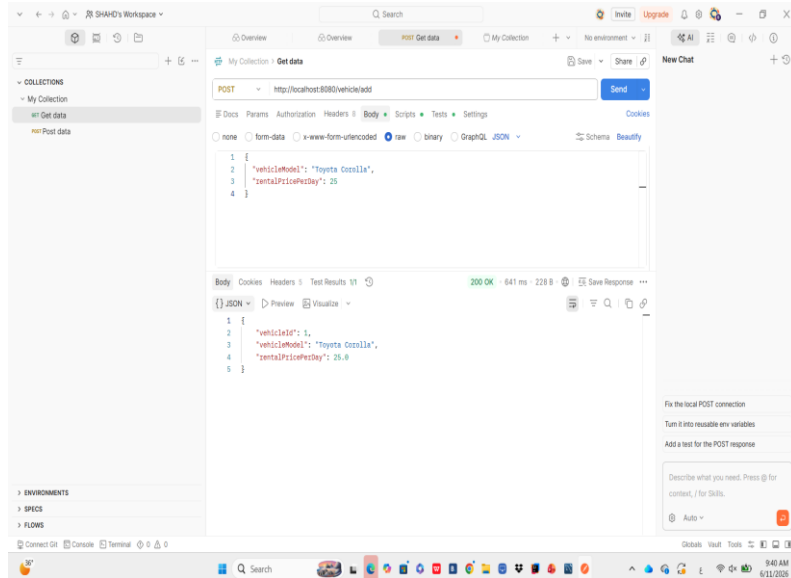
Campaign API Testing



Employee API Testing

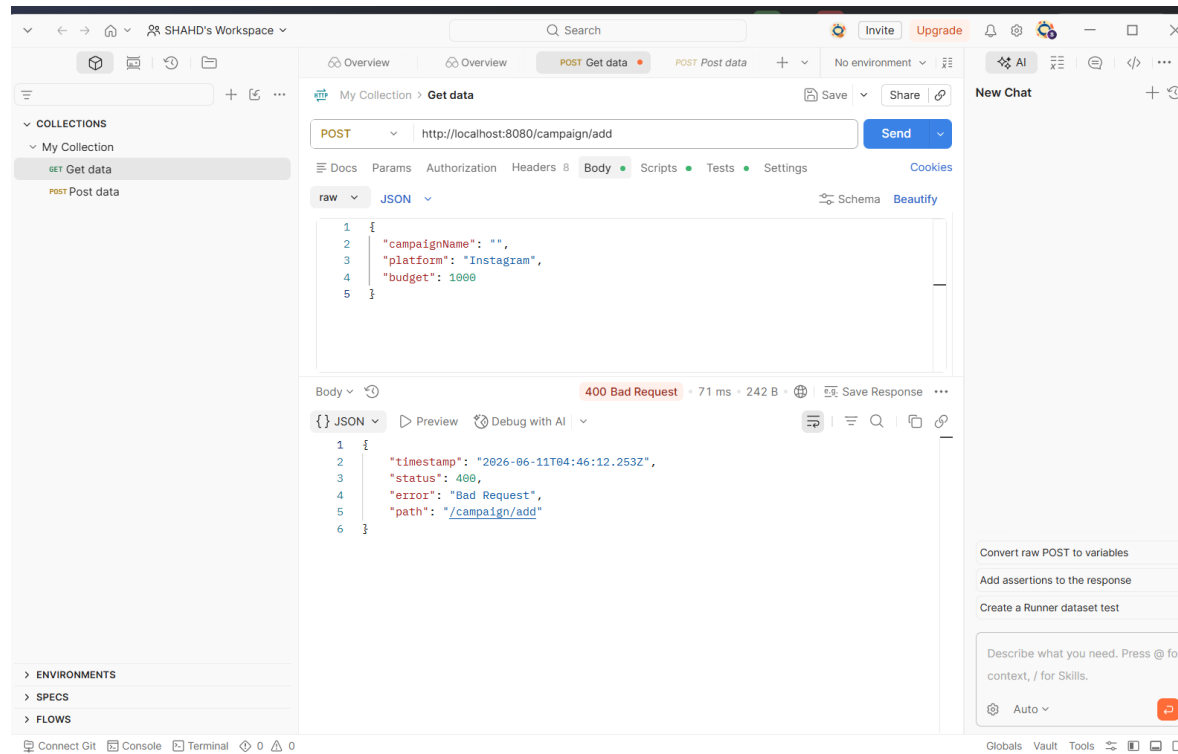


Vehicle API Testing



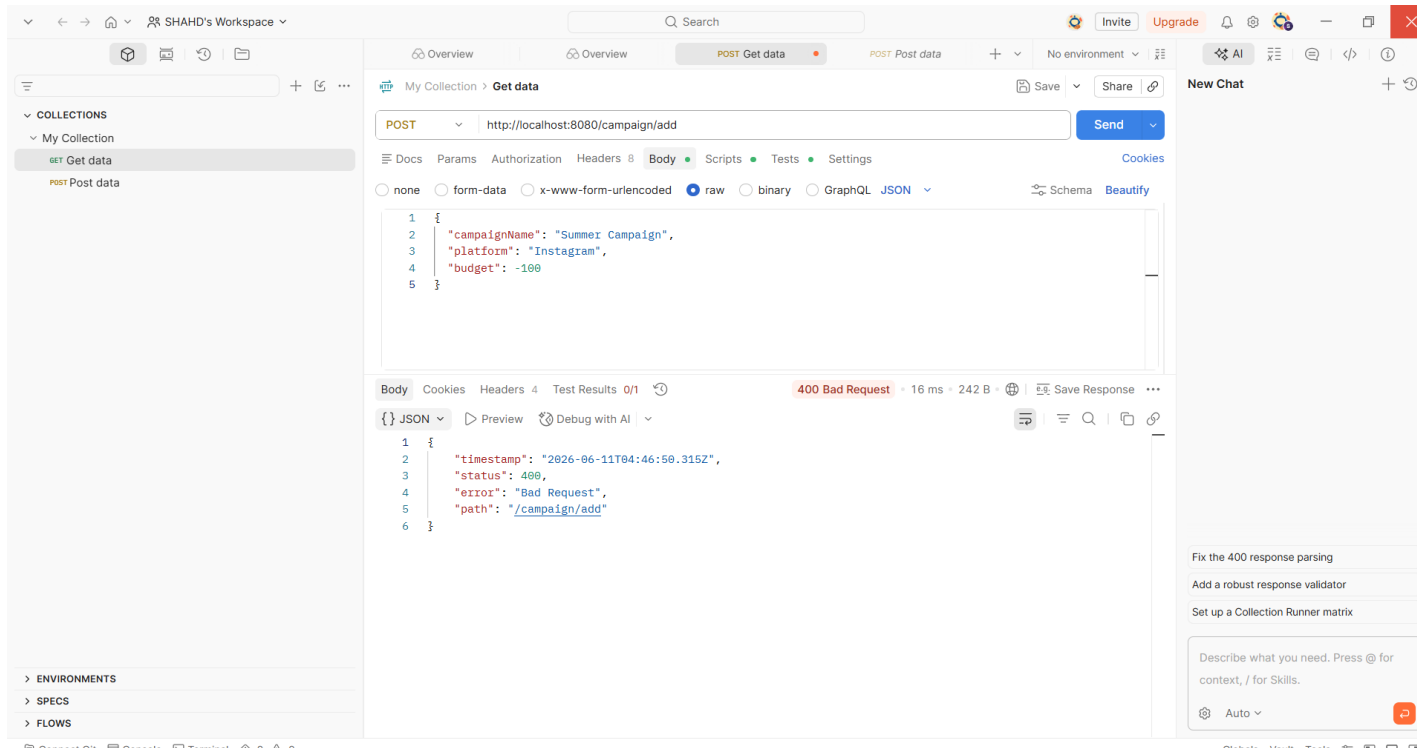
Validation Testing

Empty Name



Validation Testing

Negative Budget



MySQL Database

MySQL Workbench interface showing the 'employees' table structure and query results. The query is: `SELECT * FROM apisdadb.employees;`

Table: employees

id	department	name
1	HR	Sara
2	IT	Ahmed

Columns:

- id: int PK
- department: varchar(255)
- name: varchar(255)

Action Output:

#	Time	Action	Message	Duration / Fetch
1	08:33:23	SELECT * FROM apisdadb.employees LIMIT 0, 1000	0 row(s) returned	0.016 sec / 0.000 sec
2	09:31:40	SELECT * FROM apisdadb.employees LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench interface showing the 'campaigns' table structure and query results. The query is: `SELECT * FROM apisdadb.campaigns;`

Table: campaigns

campaign_id	budget	campaign_name	platform	employee_id
2	1000	Summer Campaign	Instagram	1

Columns:

- campaign_id: int PK
- budget: double
- campaign_name: varchar(255)
- platform: varchar(255)
- employee_id: int

Action Output:

#	Time	Action	Message	Duration / Fetch
1	08:33:23	SELECT * FROM apisdadb.campaigns LIMIT 0, 1000	0 row(s) returned	0.016 sec / 0.000 sec
2	09:31:40	SELECT * FROM apisdadb.employees LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
3	10:15:01	SELECT * FROM apisdadb.employees_campaigns LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
4	10:20:45	SELECT * FROM apisdadb.campaigns LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec